

Class-Test

1. Data Mining:

The process of extracting useful patterns and knowledge from large datasets

• Two Data Mining Tasks :-

a. classification

b. clustering

KDD process

Data collection, data cleaning, data integration, data selection, Data Transformation, Data Mining, pattern Evaluation, Knowledge presentation

Role of Data preprocessing

It improves data quality by removing errors, handling missing values, and converting data into a suitable format for accurate mining.

a. Data cleaning

The process of removing errors and inconsistencies from data.

Two data quality issues:

a. Missing values

b. noisy data

Handling missing values

• Replace with mean/median/mode

• delete records with many missing values.

Handling noisy Data

• Binning

• smoothing or clustering.

3 Data integration!

combining data from multiple sources into one dataset

challenges:

1. Data redundancy
2. Data inconsistency

Handling:

1. Data redundancy
2. Data inconsistency

Handling: -

- Remove duplicate records.
- Standardize formats and resolve conflicting values.

4 covariance

Measures how two variable change together

population covariance:

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{N}$$

• Sample covariance

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

5.

$$A = (2, 3, 5, 4, 6)$$

$$B = (5, 8, 10, 11, 14)$$

Both stock prices generally increase together

Result:-

positive covariance - the stocks rise together

6. components :

Database, Data cleaning, Database server, Data Mining Engine
Pattern Evaluation.

Working :-

Data collected, preprocessed mined for patterns, evaluated, and
finally presented to the user

7. Given

$$x = \{4, 5, 6, 7, 9\} \quad y = \{8, 3, 7, 5, 6\}$$

$$\text{covariance} = 1.53 \text{ approx}$$

Result:- Positive covariance.

8. Major preprocessing steps :

Data cleaning, Data integration, Data transformation, Data reduction,
Data discretization

These steps improve data quality before mining

9. Equal-frequency (Equi-depth) Bins :

Bin 1 : 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2 : 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3 : 35, 35, 35, 36, 40, 45, 46, 52, 70

10) Data Discretization!

converting continuous data into intervals

Equal-width binning:

All bins have the same width

Equal-frequency binning:

All bins contain nearly the same number of values

Importance:

Reduces noise and improve analysis